Vaccine availability and electricity continuity in Nigeria

FASTR Nigeria Round 1 | May to July 2025
Chika Ananaba | October 2026

Vaccine availability and electricity provision vary across Nigeria's geopolitical zones and between BHCPF and non-BHCPF facilities. The geographic priorities differ by measure: South-East recorded the highest average vaccine availability, while South-West recorded the highest electricity continuity. This report examines those comparisons using FASTR Nigeria Round 1 summaries from May to July 2025 and sets out priorities for programme review.

National vaccine availability and electricity provision

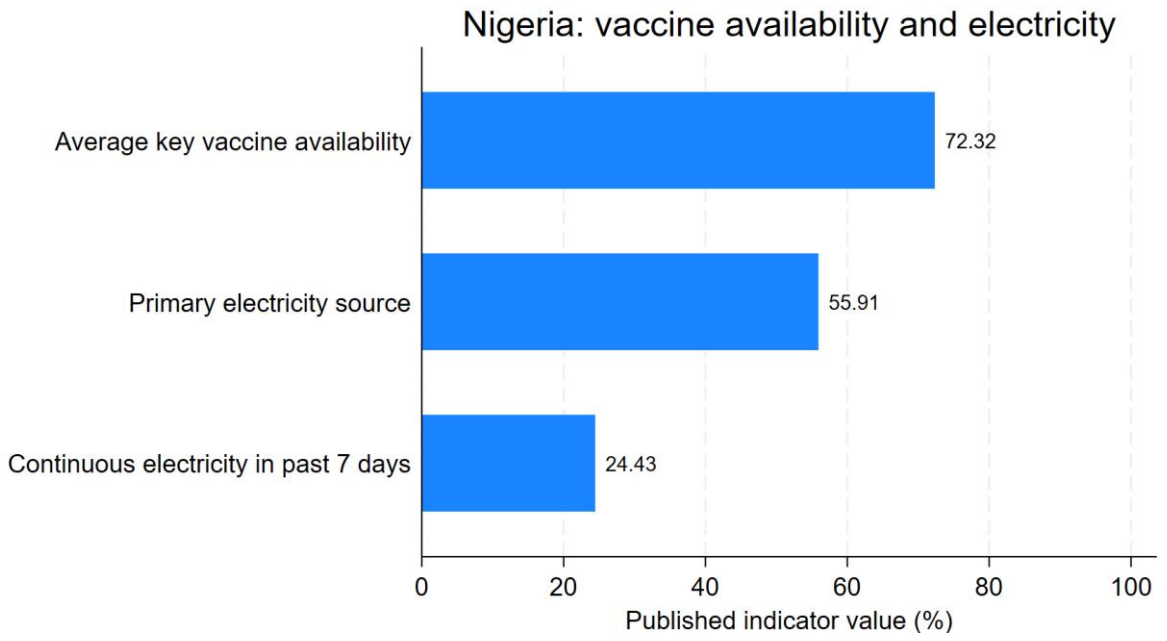


Figure 1. National vaccine availability and electricity indicators. Source: GFF FASTR, Nigeria Round 1.

Table 1. National results. Survey sample reported in the source: 3,989.

Indicator	Published value %
Average key vaccine availability	72.32
Primary electricity source	55.91
Continuous electricity in past 7 days	24.43

More facilities reported a primary electricity source than continuous electricity over the preceding seven days. The national picture therefore looks different depending on whether we examine access to a source or continuity of supply. Average availability across the selected vaccines was 72.32%.

Comparison across geopolitical zones

South-East recorded the highest average vaccine availability, while South-West recorded the highest electricity continuity. North-West ranked second for vaccine availability but last for reporting a primary electricity source. Based on the lower reported values, North-Central and North-East could be starting points for vaccine-availability reviews, while South-South and North-East could be starting points for electricity-continuity reviews.

Table 2. Results by geopolitical zone.

Zone	Vaccine average %	Electricity source %	Continuous electricity %	Survey sample
North-Central	68.36	56.56	26.98	720
North-East	69.55	48.17	18.86	518
North-West	75.78	42.56	24.81	869
South-East	81.11	68.87	23.99	610
South-South	74.34	59.91	18.22	572
South-West	70.13	70.38	32.31	700

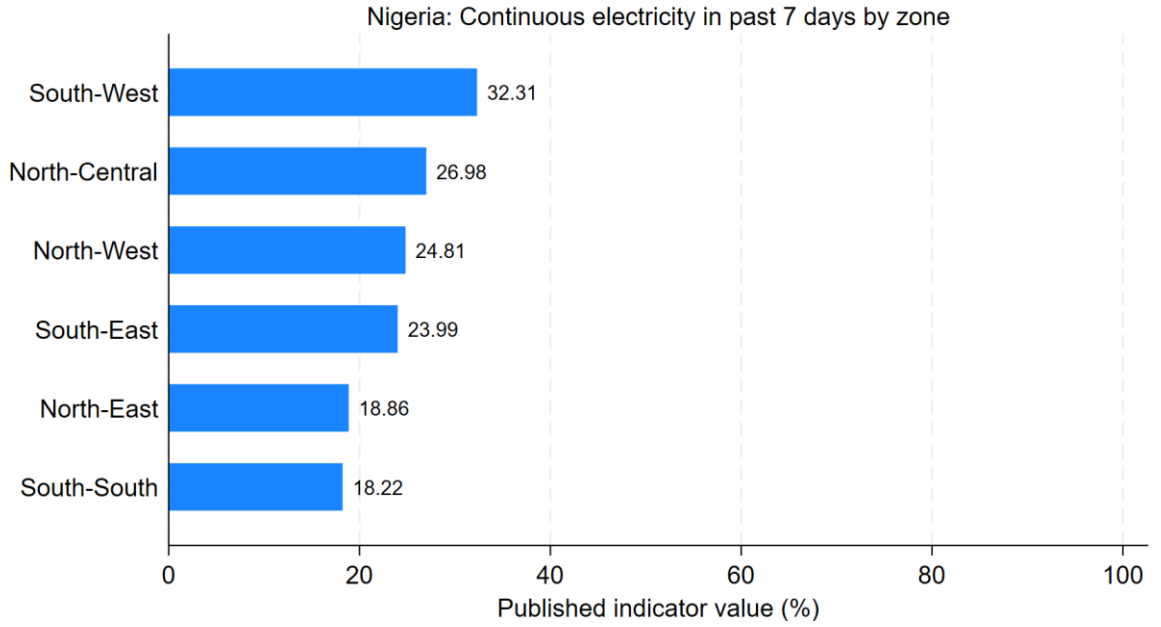


Figure 2. Facilities reporting continuous electricity during the preceding seven days. Source: GFF FASTR, Nigeria Round 1.

Fewer than one-third of facilities in every zone reported continuous electricity over the preceding seven days. South-South and North-East recorded the lowest continuity values. North-Central and North-East recorded the lowest average vaccine availability. North-East may therefore warrant closer review of both vaccine availability and electricity continuity.

Supporting vaccine availability and electricity-source charts appear in Appendix A.

Comparison by BHCPF participation

Facilities participating in the Basic Health Care Provision Fund (BHCPF) recorded higher values across all three measures. The largest difference was in reporting a primary electricity source, at 37.99 percentage points. The differences in vaccine availability and electricity continuity were smaller, at 15.13 and 17.93 points respectively.

Table 3. Results by BHCPF participation. Survey samples: non-BHCPF 828; BHCPF 3,161.

Indicator	Non-BHCPF %	BHCPF %
Average key vaccine availability	67.43	82.56
Primary electricity source	43.64	81.64
Continuous electricity in past 7 days	18.64	36.56

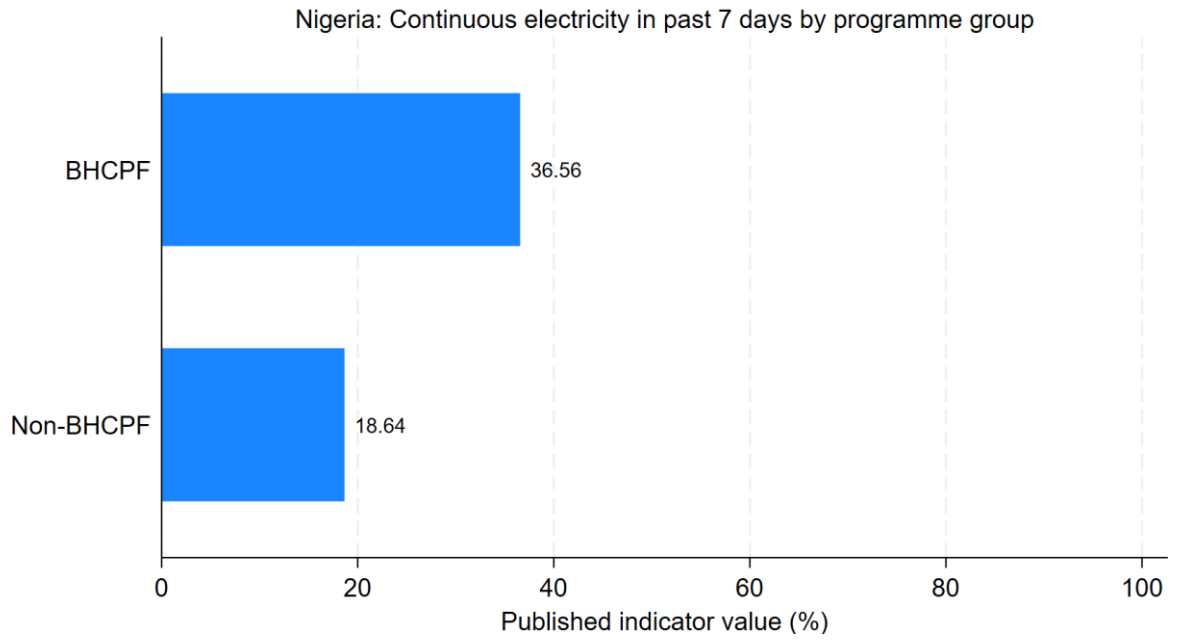


Figure 3. Electricity continuity by BHCPF participation. Source: GFF FASTR, Nigeria Round 1.

Continuous electricity was reported by 36.56% of BHCPF facilities and 18.64% of non-BHCPF facilities. BHCPF facilities had the higher figure, but fewer than half of facilities in either group reported continuity

Supporting vaccine availability and electricity-source charts appear in Appendix B.

Priorities for further investigation

South-East recorded the highest average vaccine availability, while South-West recorded the highest electricity continuity. Based on the lower reported values, North-Central and North-East could be starting points for vaccine-availability reviews, while South-South and North-East could be starting points for electricity-continuity reviews.

Vaccine availability

National Primary Health Care Development Agency (NPHCDA) and state immunisation teams could examine availability by individual vaccine in North-Central and North-East, alongside stock and delivery records. This could help identify which products and facilities account for the lower zone averages and guide supply follow-up.

Electricity provision

State Primary Health Care Boards and Agencies, working with local government health authorities and facility managers, could investigate electricity continuity in South-South and North-East. Information on interruptions, power sources and equipment could help establish what support is appropriate.

BHCPF comparison

The NPHCDA gateway and state BHCPF implementation teams could compare facilities with similar locations and characteristics to examine whether the reported differences persist. Information on funding received and its timing could support further analysis of how facility conditions relate to programme support.

NPHCDA describes its BHCPF gateway as operating through State Primary Health Care Boards/Agencies, local government health authorities and primary health centres: https://nphcda.gov.ng/bhcpf/.

Size of the observed differences

Table 4. Differences within each indicator, calculated from full-precision source values.

Indicator	Zone minimum %	Zone maximum %	Zone gap pp	BHCPF gap pp
Average key vaccine availability	68.36	81.11	12.75	15.13
Primary electricity source	42.56	70.38	27.83	37.99
Continuous electricity in past 7 days	18.22	32.31	14.09	17.93

pp = percentage points. Zone gap is highest minus lowest; BHCPF gap is BHCPF minus non-BHCPF.

Reporting a primary electricity source had the widest range across zones and the largest BHCPF versus non-BHCPF difference. This was the measure on which the reported groups differed most.

Methods and reproducibility

The source was GFF FASTR's published summary workbook, FASTR-Data.xlsx, country-round NG_R1, with a May–July 2025 survey period. Vaccine availability is the published average across selected tracer vaccines; the electricity indicators are percentages of facilities reporting a primary source or continuous electricity over the preceding seven days. The Stata 19 do-file selects SUP_05AAA_*, INF_04 and INF_05 from the national, region and facility-type sheets, converts numeric values, reshapes subgroup tables and merges indicator metadata using explicit keys. National results are taken directly from the published national summaries.

Checks cover country-round and subgroup headers, survey period, unique indicator and subgroup keys, expected output sizes, complete numeric values, percentage ranges and positive integer survey samples. Zone and programme samples each reconcile to 3,989 per indicator. The run completed on 13 October 2026 without Stata error codes, and the CSV tables were checked against their saved DTA datasets. The report uses the original Stata charts, with repeated footer notes omitted from their display; chart values, labels and scales are unchanged.

Limits of the evidence

The results are descriptive summaries from one survey round. Reported survey samples are not confirmed indicator denominators and were not used as weights. Confidence intervals and significance tests were not calculated. Separate subgroup summaries do not identify whether vaccine and electricity gaps occur in the same facilities, and BHCPF comparisons do not estimate programme effects. Vaccine availability differs from vaccination coverage, and general electricity measures do not establish cold-chain performance. The proposed follow-up requires vaccine-specific or facility-level information.

Source: GFF FASTR data portal, https://data.gffportal.org/key-themes/FASTR; workbook FASTR-Data.xlsx. Tables: readiness_national_table.csv, readiness_zone_table.csv, readiness_programme_table.csv and readiness_comparison_gaps.csv. Audit trail: FASTR_Nigeria_expanded.log and readiness_extraction_checks.txt. Code: FASTR_Nigeria_expanded.do.

Appendix A Supporting zone charts

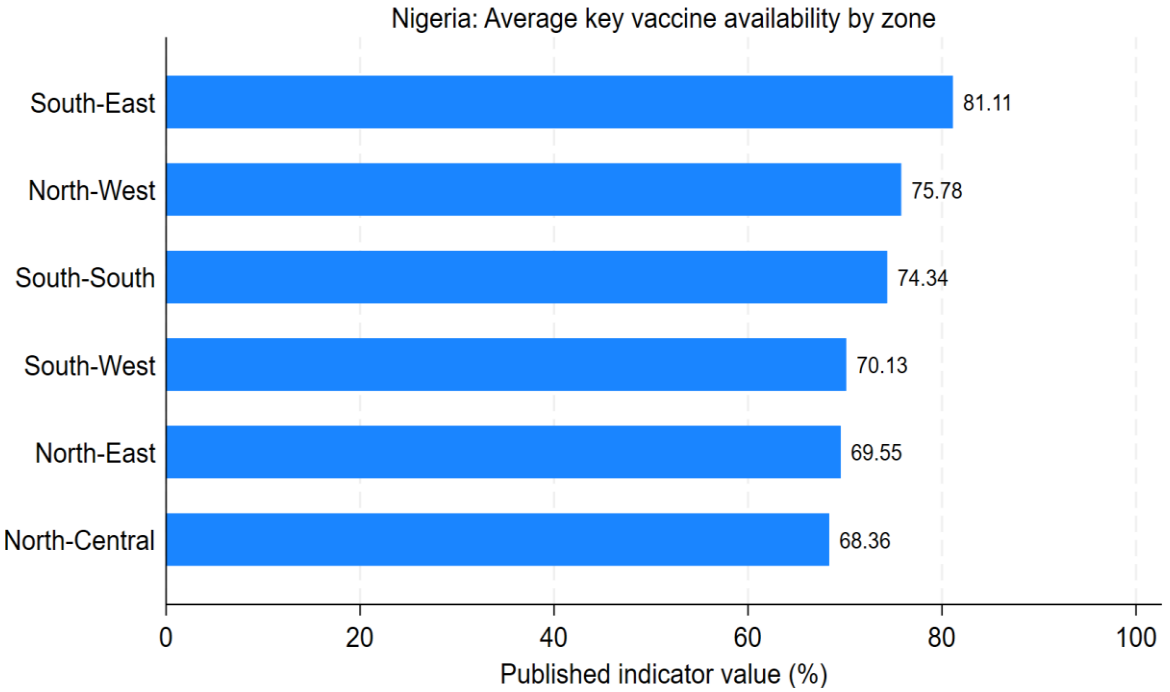


Figure A1. Average key vaccine availability by zone. Source: GFF FASTR, Nigeria Round 1.

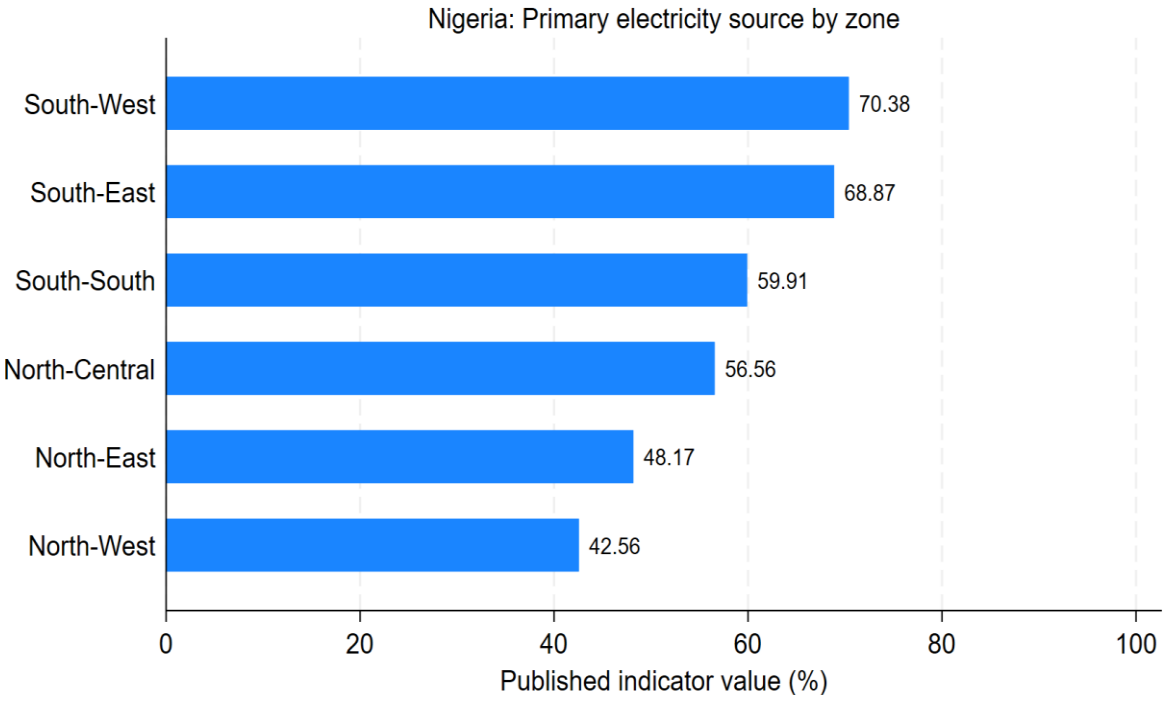


Figure A2. Facilities reporting a primary electricity source by zone. Source: GFF FASTR, Nigeria Round 1.

Appendix B Supporting programme charts

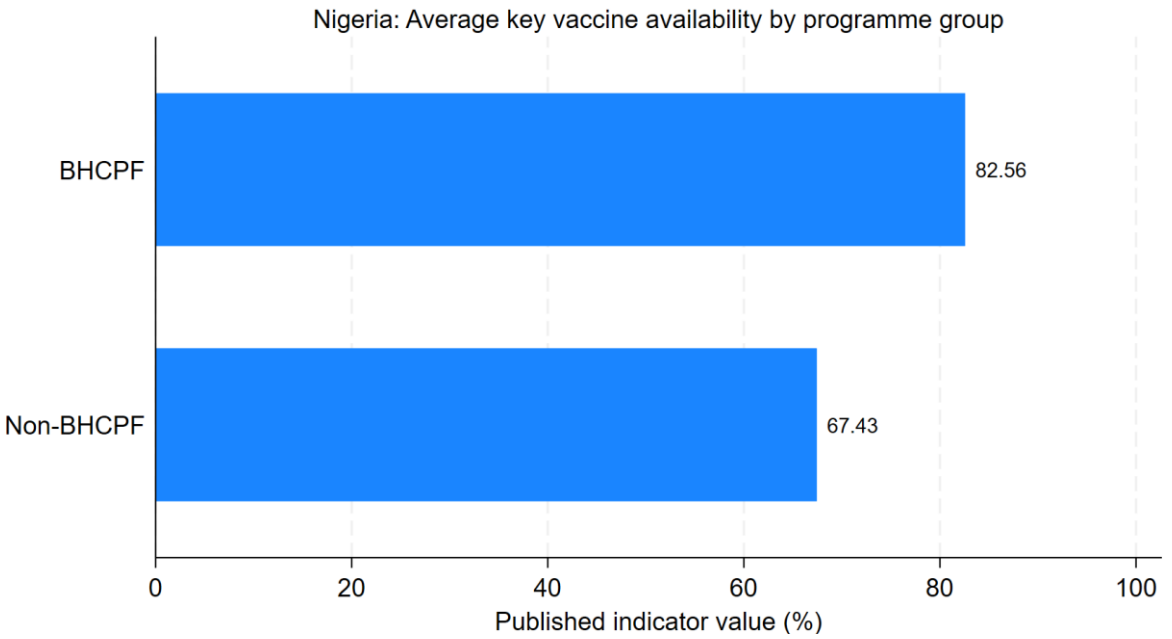


Figure B1. Average key vaccine availability by BHCPF participation. Source: GFF FASTR, Nigeria Round 1.

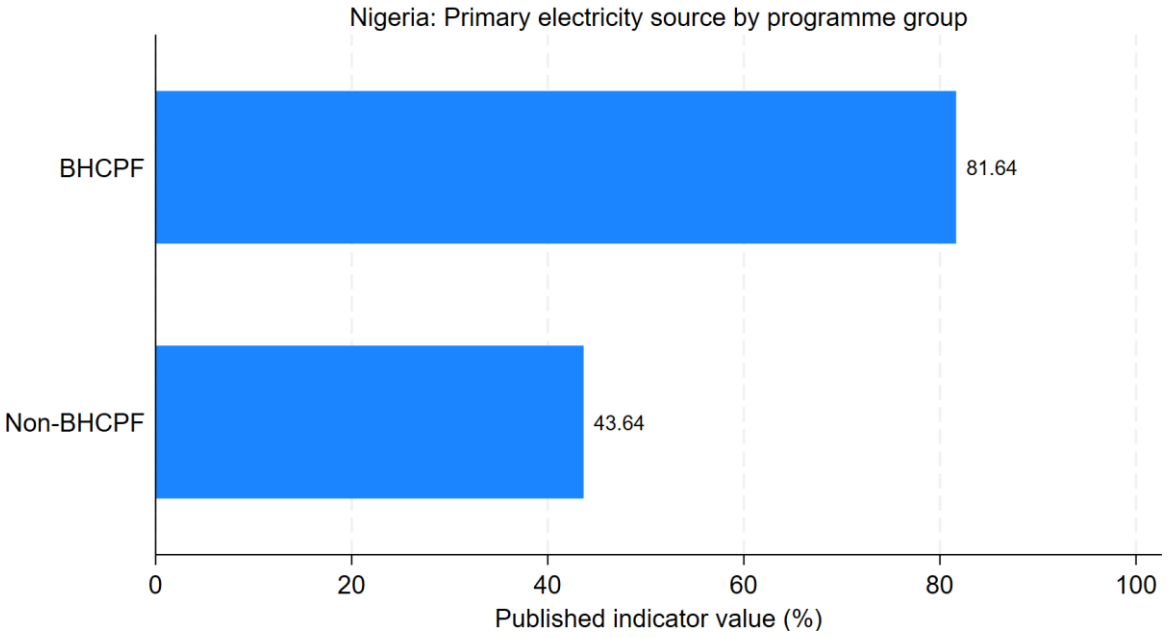


Figure B2. Facilities reporting a primary electricity source by BHCPF participation. Source: GFF FASTR, Nigeria Round 1.